Petra Zlatarić-Novosel

Machine Learning Engineer | NLP and Document AI

Professional Summary

Machine learning engineer with production experience in text classification, named entity recognition and model monitoring on Azure. Fine-tuned transformer models for document processing at approximately 15,000 documents per day. Earlier banking experience includes customer complaint classification and churn prediction. Doctoral research focuses on named entity recognition in Croatian legal texts.

Technical Skills

- Languages and data: Python, SQL, pandas, PostgreSQL, Oracle SQL.
- NLP and model development: PyTorch, Hugging Face Transformers, spaCy, scikit-learn, fastText, TF-IDF, XLM-R, BERTic.
- Production serving and MLOps: FastAPI, Docker, Azure App Service, Azure ML, Azure Functions, MLflow, Evidently.
- Text datasets and retrieval: Label Studio, annotation guidelines, sentence-transformers, FAISS; RAG prototype experience.

Professional Experience

ML Engineer | Slavonica Document AI | 2023-Present

- Develop document classification and named entity recognition models for Croatian public administration documents.
- Fine-tuned XLM-R base to classify OCR-processed invoices, contracts and decisions into 23 classes, achieving 94% accuracy and macro F1 of 0.91.
- Served the classifier through FastAPI in Docker on Azure App Service, processing approximately 15,000 documents per day.
- Fine-tuned BERTic to identify persons, organisations, OIB numbers, addresses and legal references in Croatian administrative text.
- Built an annotated dataset of 4,000 documents with annotation guidelines and two annotators, achieving inter-annotator kappa of 0.87. This work was published at a regional workshop in 2024.
- Implemented weekly model drift reporting on input text length, vocabulary and predicted class distribution, with alerts when PSI exceeded 0.2, using Python, Evidently, Azure Functions and PostgreSQL.
- Detected a silent client-side OCR change that reduced accuracy by 9 percentage points; the issue was fixed within a week.

Data Scientist | Baranjska kreditna banka | 2021-2023

- Developed models for customer churn prediction and customer complaint classification.
- Built an 11-topic complaint classifier using a TF-IDF and logistic regression baseline, followed by fastText, for weekly reporting to customer service management.
- Used scikit-learn, pandas and Oracle SQL for complaint classification work.

Selected Personal Project

Retrieval-augmented Question Answering Demo | 2025

- Built a weekend RAG prototype for question answering over Croatian laws using sentence embeddings, FAISS and an open-weight LLM accessed through an API.
- Personal prototype only; assessed through manual checks, not deployed to production.

Education

Doctoral studies, in progress | Sveučilište Drava
Expected completion: 2027
Research topic: Named entity recognition in Croatian legal texts.

Certification

Microsoft Certified: Azure AI Engineer Associate | 2024

Teaching and Languages

- Delivered a 90-minute guest lecture at an NLP summer school in 2025.
- English: used daily in meetings with a client from Slovenia.